GRAVITYSORT · DESIGN DECISIONS REGISTER

GravitySort design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: fixes F1 to F6 made under Amish's standing instruction of 2026-10-23 (STEM check, independent review and detailed CAD); open decisions O1 (concept selection, GVS-DDR-004), O2 (containment band), O3 (tank sway), O4 (spindle supports) and O5 (fluidization flow) added, Proposed, awaiting Amish; items to confirm and value engineering updated (USD 594)
0.7	2026-10-03	Amish Chadha	Open item 1 decided by Amish on 2026-10-03 (option a: tank post on the centre line, braced, 10 degree any-direction tip criterion); moved to decisions made
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried into the model, BOM, calculations and build plan; one open decision added (tank on the centre line and the tipping criterion, made to carry out decision 1); Value engineering updated to USD 580
0.5	2026-10-02	Amish Chadha	

			Amish approved the recommendations for open decisions 1 to 5; moved to decisions made
0.4	2026-10-01	Amish Chadha	Amish accepted the change made while adding the bump stop (slotted pitman pin, leaning legs; GVS-DDR-003 Table 2); added to decisions made; no open item changed
0.3	2026-10-01	Amish Chadha	Table bump stop now in the model and build plan (follow-up note removed); buffer added to the items to confirm; Value engineering updated to USD 569
0.2	2026-10-01	Amish Chadha	Amish accepted the recommendations of open items 1 to 4, 9, 10 and 12 (GVS-DDR-003 accepted; R11 restated); moved to decisions made; open items renumbered 1 to 5
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from the review notes, the decision records and the design for construction

GravitySort design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`, GVS-BLD-001) describes the design as it stands and does not list open decisions.

Open decisions

Each is Proposed, awaiting Amish.

#	Decision	Options	Recommendation	Record
O1	Concept selection	(A) fluidized bowl and shaking table, the current design; (B) diaphragm jig and shaking table; (C) carpeted sluice and gold pans. Weighted scores: C 172, A 165, B 157; all three are within 10 % of the top score, so under STANDARDS section 21 all go to Amish without a pick	No pick (within 10 %). If UR-003, fine gold, is treated as a gate (a score of 3 or more), only A passes (B scores 2, C 1). Under section 21 the current design A stays in the repo unless Amish decides otherwise	GVS-OPT-001; GVS-DDR-004
O2	Containment of a liner fragment (PR-016)	(a) keep the 6 mm HDPE tub and 10 mm lid, and make an impact test the first TRL 4 item, before any spin test; (b) fit a 3 mm steel band, about 200 mm tall, round the tub at the bowl's height now (about 6.4 kg, which takes the mass to about 106 kg, over PR-011's 100 kg)	(a). The calculation shows containment face first (57 J against 40 J at 1,200 rpm) but not edge first (14 J against 22 J at 900 rpm) nor for a quarter of the whole bowl group (125 J, the independent review's bounding case); Amish's rule of 2026-10-01 (band only if the check fails) cannot be applied cleanly because the result depends on how the fragment strikes, and (b) breaks a requirement he restated	GVS-CAL-001 section 10; GVS-STM-001 S10
O3	Sway of a full tank near the table drive frequency (STEM S33)	(a) raise the brace node from 1.35 to 1.6 m (about 7.5 Hz instead of 3.5 Hz; braces about 0.25 m longer each, about +0.5 kg, which takes PR-011 to about 100.1 kg) and take one of the savings in Value engineering to stay under 100 kg; (b) a rule to run the table only with the drum drained, wash water from a bucket (no mass); (c) both	(a) with the bronze-bush saving on the table head, plus (b) as an interim rule. Both change a decided layout or a requirement margin, so they are Amish's	GVS-CAL-001 section 11; GVS-STM-001 S33
O4	Spindle supports and the first critical speed (STEM S8)	(a) keep the frame and check the critical speed by a tap test on the first frame (TRL 4), stiffening the spindle supports only if it is under 1,500 rpm; (b) stiffen now with a cross member under each bearing plate (mass over PR-011)	(a): 1,274 to 1,976 rpm on paper, always above the 900 rpm limit	GVS-CAL-001 section 10
O5	Fluidization flow and pressure at the low end of commercial practice (STEM S3)	(a) keep 8 to 15 L/min and sweep the flow in the TRL 4 recovery test; (b) design for more flow now (more head or a pump, against the 2026-10-02 decision; more water against PR-008)	(a)	GVS-CAL-001 section 6; GVS-STM-001 S3

Fixed under Amish's standing instruction of 2026-10-23

Amish, 2026-10-23: when a STEM check row fails, propose and implement the best fix for the design and its constructability in the same session and record it here as decided under that instruction; a fix that changes the working principle, pitch or scope goes to Amish instead. Amish can reverse any of these.

#	Failed check	Options weighed	Fix made (2026-10-25)	Cost and mass	Record
F1	STEM S20: the header drum was drawn 380 mm across and 400 mm tall, which holds 45 L, not the 60 L the design, the decision of 2026-10-02 and the CAL use	(a) draw the bought 60 L drum at its real height, 560 mm; (b) call it a 45 L drum and keep the geometry	(a): `tank_z` 1,700 to 2,260 mm. Mid-tank level rises 80 mm (+0.8 kPa at the union); the machine still stands on 13.1 degrees with a full tank (was 13.5; criterion 10); overall height 2.26 m (was 2.10 m)	None (same bought drum)	GVS-CAL-001 sections 6 and 11
F2	STEM S24: the fluidization water reached the 1.0 mm holes unfiltered from a recirculating pond; commercial fluidized concentrators need clean fluidization water (Veiga and Gunson 2020)	(a) screen filter on the hose; (b) a settling drum before the header tank; (c) larger holes (changes the fluidization design)	(a): 3/4 in screen filter with a flush valve and a 120 mesh stainless screen, on the hose's drop below the rotameter. Loss 0.75 kPa clean, 1.49 kPa at the cleaning point; lowest net ring pressure stays positive (7.0 kPa with the climb to the rings counted)	+$14 (DIG P13-120, BOM line 13 now $55); +0.3 kg	GVS-CAL-001 section 6; BOM line 13
F3	STEM S16: the BOM asked for a buffer rated 600 N, against a peak of 594 N at the 6 mm setting; GVS-CAL-001 section 9 had already chosen 800 N	(a) specify 800 N or more, as the CAL chose; (b) limit the setting range to 4 mm	(a): BOM line 22 now 800 N or more	None expected	BOM line 22
F4	STEM S21: the gold balance used 8 h of feed (1.6 t, 8.0 g of gold) while the throughput restated by Amish on 2026-10-02 is 1.55 t with three flush stops	(a) use the feed time of 7.75 h	(a): 7.75 g in, 4.7 g smelted per shift on the reference ore; overall 60 % unchanged	None	GVS-CAL-001 section 13; `media/flow.png`
F6	STEM S4 (found by the independent review): the supply calculation left out the climb from the union to the rings (4.5 to 5.6 kPa); with it the lowest ring had 6.0 kPa net at 600 rpm against the proposed 10 kPa on which the 1.7 m post was decided	(a) 3/4 in bores for the ball valve and rotameter, cutting the fittings loss; (b) raise the post again (reverses part of the 2026-10-02 decision and worsens tipping); (c) a pump on the fluidization line (Amish decided against it)	(a): ball valve full bore and rotameter with 3/4 in ports; fittings loss 7.5 to 3.4 kPa; 12.6 kPa at the union and 10.0 kPa at the lowest ring at mid-tank, 7.0 kPa in the worst case	Rotameter price to confirm (BOM line 13 kept at $55)	GVS-CAL-001 section 6; BOM line 13
F5	Detailed CAD (D3): the nuts of the head pillow block at y 40 struck the shelf gusset at y 50	(a) move the gusset to y 59; (b) move the pillow block	(a): gusset centred at y 59, still under the shelf (which ends at y 65)	None	`cad/src/model.py`; GVS-DWG-116

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The two flanged bearing units (UCF205 class): 95 mm square flange, bolts 70 mm apart, 25 mm bore with set screws	They set the holes in the two bearing plates	GVS-DDR-003, P2
2	The shaft flange hub: 25 mm bore, 80 mm flange, four holes on a 60 mm circle, room for a 6 mm cross pin	It joins the bowl and jacket to the spindle	GVS-DDR-003, P4
3	The rotary union: 1/2 in BSP on the shaft side, side water port, seal drag near 0.10 N m	Pedal power (R6) assumes this drag; the nipple welded into the spindle must match its thread	GVS-CAL-001 sections 7 and 10
4	The right-angle 1:1 gearbox: through input shaft about 33 mm above its feet, feet about 100 x 80 mm, output shaft long enough for the drive pulley 85 mm below the feet	It sets the jackshaft height and the member spacing	GVS-DDR-003, P9; GVS-PRC-001 open questions
5	The four 20 mm pillow blocks (UCP204 class, centre height 33.3 mm, bolts 95 mm apart)	Two set the jackshaft height on the gearbox members; two carry the table head shaft	GVS-DDR-003, P8 and P14
6	The two rubber pipe grommets for 63 mm and 75 mm pipe that seal in a 6 mm curved wall	They seal the standpipe and tailings pipe without welding	GVS-DDR-003, P6
7	The 430 mm HDPE drum: flat floor, 6 mm wall, food-grade or clean	It is the splash tub	GVS-DDR-003, P6
8	The used bicycle parts: a bottom bracket shell that can be cut from a scrap frame and welded, crank axle length, a 27.2 mm seat post	They set the outrigger's pedal post and seat tube	GVS-DDR-003, P8
9	The disc caliper's mount (post mount or the older standard) and the disc's 6-bolt pattern	They set the caliper bracket holes and the brake flange	GVS-DDR-003, P15
10	MotionCore: the reference hub motor's no-load speed (assumed 250 rpm), a sprocket on its disc mount, axle length for dropouts 76 mm apart inside, a speed limit that maps to 900 rpm at the spindle, a brake input for the lid switch	They set the 1.2:1 step-up, the cradle and the motor speed cap	GVS-DDR-002, item 14; review note 2026-09-25
11	The bump stop's rubber buffer: 40 mm across, 30 mm long, about 55 Shore A, M8 female thread, rated 800 N or more in compression	The stop's force, squeeze and setting range are sized for this buffer (GVS-CAL-001 section 9)	GVS-DDR-003, A2; F3
12	The 60 L header drum: about 380 mm across and 560 mm tall, with a bung that takes the 3/4 in valve near its bottom	The drum height sets the head at the union and the tipping check	F1
13	The 3/4 in screen filter: about 0.75 kPa loss at 12 L/min when clean, a flush valve, 120 mesh or coarser up to 0.3 mm	The supply has no pressure to spare at mid-tank for the proposed 10 kPa at every ring	F2

Value engineering

Value-engineering target: USD 455 (a hypothetical control target, not a limit; Amish, 2026-10-01). Estimated cost of the constructable design: USD 594 (USD 139 over the target), excluding the MotionCore kit, reference motor and battery; the screen filter of F2 added $14 on 2026-10-25. Main cost drivers and savings worth trying:

- **Main cost drivers.** The jackshaft, gearbox, chains and pulleys ($60); the jacket, hub and rotary union ($55); the water supply ($55, with the filter); the table stand and head ($52); the frame ($47); the bowl ($46). Making the design buildable added $103, mostly the table head and tensioner, more frame tube and plate, and the tub fittings (GVS-DDR-003); the table bump stop adds $11 more (bracket and striker $6, buffer and fixings $5); the decisions of 2026-10-02 add $11 (the braced 1.7 m tank post $4, the flush container with its hasp $7).
- **A quarter-turn belt instead of the bevel gearbox,** about $25 less (GVS-DDR-001, item 1). Worth a test, since a used gearbox is also the hardest part to find.
- **Used bearing units and pillow blocks** from scrap farm or factory machinery, about $10 to $15 less across the six bearings.
- **Run the table head shaft in two plain bronze bushes** in the head plate instead of two pillow blocks, about $8 less; the shaft turns slowly (about 300 rpm).
- **Take the table wash water from the main hose** with a tee at the rotameter instead of a separate branch and valve, about $4 less.
- **Buy steel as offcuts** and use one plate thickness (5 mm) throughout, a few dollars and a little mass.
- Not recommended: the non-fluidized bowl (about $45 less) or building the table later (about $100 deferred), because both change what the machine does.

Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items: two stages, fluidized bowl with rotary union, cast PU liner, pedal baseline with the MotionCore option and one time-shared drive, battery left to the user, borax smelting outside the machine, partner screening criterion	Amish: "i accept all your recommendations, go with them across all repos."	GVS-DDR-001, GVS-DDR-002
2026-09-25	Budget from $350 to $450; R12 restated for the pedal drive; frame in 25 x 25 x 1.5 mm tube; 3/4 in fluidization hose and graded holes; disc brake with parking latch and lid interlock pin, speed display, 1.2:1 motor step-up, MotionCore lid switch	Amish, same instruction	GVS-DDR-002, items 1, 9, 10, 11a and 14
2026-09-26	Budget set to $455 to cover the priced BOM	Amish: "i approve all the budget items."	GVS-DDR-002 v0.2
2026-09-26	GravitySort chosen for the first batch of product renders	Amish	`docs/REVIEW.md`, session of 2026-09-26
2026-09-30	Make the design physically buildable while drawing the build plan; the changes are recorded in GVS-DDR-003 and were accepted on 2026-10-01 (below)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	GVS-DDR-003
2026-09-30	Open decisions are kept out of the build plan, in this register	Amish: "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	`.kit/STANDARDS.md` section 18
2026-10-01	The budget is a value-engineering target, not a limit; cost is reported against it	Amish: "the budgets are a hypothethical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	`.kit/STANDARDS.md` section 18
2026-10-01	Design for construction accepted: the changes P1 to P15 and their knock-on changes, as made	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, Tables 1 and 2
2026-10-01	R11 restated: total 100 kg or less, every load 30 kg or less (was 80 kg in all); the 95.7 kg design meets it on paper. Follow-up: try the savings of about 16 kg (option b) at TRL 4	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, A1; GVS-REQ-001 v0.7
2026-10-01	Asymmetric table stroke: an adjustable rubber bump stop at the return end of the stroke for the prototype. Follow-up: a toggle head (option b) only if the first test shows the stop is not enough	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, A2

Date	Decision	Decided by	Record
2026-10-01	Pedal position: keep the bicycle-style seat, about 910 to 960 mm up, seat post adjustable. Follow-up: check with two riders at TRL 4	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-DDR-003, A3
2026-10-01	Containment of a liner fragment: check by calculation before any spin test; add a steel band round the tub at the bowl lip height only if the check fails	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	GVS-CAL-001 section 10; review note 2026-09-25
2026-10-01	Lid: keep the plain 10 mm HDPE lid, no sight window	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	Review note 2026-09-26, item 4
2026-10-01	Renders: the cranks turned 100 degrees from top dead centre is a render pose only; the model keeps its crank angle	Amish: "i agree with your recommendations for both GrowRider and GravitySort"	Review note 2026-09-26, item 1
2026-10-01	Table bump stop, change made to let the deck strike the stop: the pitman pin works in a 12 x 20 mm slot in the deck cheeks (8 mm of free play), and the plywood legs are set leaning about 5 mm so they press the deck on the buffer with about 200 N. Accepted as made; no design change	Amish: "I approve of your recommendations for PicoFlow and GravitySort"	GVS-DDR-003, Table 2 (v0.4)
2026-10-02	Head for the fluidization supply: raise the header tank post to 1.7 m (option a), braced so a full 60 L tank cannot tip it; no pump on the fluidization line. Carried into the model, drawings, build plan, BOM and GVS-CAL-001 v0.8 on 2026-10-02 (8.6 kPa at the union at mid-tank)	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 11b
2026-10-02	Water at pedal-only sites: gravity supply from upstream is the site rule for the first field trial; where that is impossible, a bought treadle or hand pump worked in turns by the crew, or a small 12 V pump at sites with the MotionCore battery	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 12
2026-10-02	R2 restated as 200 kg/h of feed time, about 1.55 t in an eight-hour shift with three flush stops; the design feed stays at 200 kg/h	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 13
2026-10-02	First co-design partner and country: Colombia, where the mercury ban is in force, with a miners' cooperative introduced through the Alliance for Responsible Mining in Medellin as the first candidate to approach	Amish: "i approve your recommendations for all 555 open decisions."	GVS-DDR-001, item 8b
2026-10-02	Concentrate security: padlock hasps on both the concentrate box and the flush container		GVS-PRC-001, open questions

Date	Decision	Decided by	Record
	as the baseline, and a two-person rule for opening them proposed to the partner; both to be confirmed with the partner. Flush container with hasp added to the model and BOM (item 23) on 2026-10-02	Amish: "i approve your recommendations for all 555 open decisions."	
2026-10-03	Open item 1, option (a): the 1.7 m tank post stands on the centre line and is braced; "a full tank cannot tip it" means a 10 degree slope in any direction with a full tank (tips at 13.5 degrees, 193 N side push at the tank, better than the earlier 1.25 m layout at 11.6 degrees); no wider feet or outriggers	Amish: "GravitySort - i accept your recommendation"	GVS-CAL-001 v0.8 section 11; REVIEW.md, session 2026-10-03